

AziCarpool Application - Requirements Document

Purpose: To define clear, comprehensive functional and non-functional requirements for an internal employee carpooling application. This document serves as a reference for developers using code automation tools (e.g., GitHub Copilot) and for QA to generate automated unit tests and functional test cases.

Key Features & Objectives

- Employees can create carpool rides from their location to office or from office to their location at specific times.
- Employees can search and join existing carpool rides based on route compatibility.
- Authentication via company email and OTP verification.
- Map integration for visualizing routes and estimating travel times.
- Ride owners can approve or reject join requests from other employees.
- Seat availability tracking as employees join carpools.
- Automated notifications for ride status updates and reminders.
- Test coverage using code automation tools for both development and QA workflows.

User Roles

Role	Description
Employee	Can create rides as drivers or join rides as passengers.
Ride Owner	Employee who created the carpool ride and manages join requests.
Admin	Manages system settings, user verification, and resolves disputes.

Access Control

- All employees can view available carpool rides.
- Only verified employees with complete profiles can create or join rides.
- Ride owners have exclusive rights to approve/reject passenger requests.
- Admins can moderate all rides and user activities.

System Entities

Entity	Attributes
User	ID, Name, Email, Phone, Department, Home Location, Verification Status
Vehicle	ID, Owner (User ID), Make, Model, Color, License Plate, Capacity
Carpool Ride	ID, Driver (User ID), Vehicle ID, Starting Location, Destination, Date, Departure Time, Available Seats, Route, Status

Entity	Attributes
	(Active/Completed/Cancelled)
Ride Request	ID, Ride ID, Requester (User ID), Pickup Location, Status (Pending/Approved/Rejected)
Location	ID, Address, Latitude, Longitude
Notification	ID, User ID, Message, Type, Read Status, Timestamp

Functional Requirements

1. User Registration & Authentication

- Employees sign up using company email only.
- Authentication via OTP sent to company email.
- Complete profile with personal details and optional vehicle information.
- Admin verifies employee accounts.

2. Ride Creation

- Ride owners can create carpool rides specifying:
 - Starting location (home or office)
 - Destination (office or home)
 - Date and time of departure
 - Available seats
 - Any preferences or notes

3. Ride Discovery

- Map-based search for available rides.
- Filter by time, proximity to route, available seats.
- View ride details and driver information.

4. Ride Joining Process

- Request to join with pickup location.
- Ride owner receives notification of join request.
- Owner can approve or reject requests.
- Seat count decreases with each approval.

5. Map Integration

- Visual display of ride routes.
- Calculation of ETAs and distance.
- Suggested pickup points optimization.

6. Notifications System

- Request status updates.
- Ride reminders for both drivers and passengers.
- Cancellation alerts.

- Real-time updates for delays or changes.

7. Ride Management

- Drivers can track passengers.
- Update ride status (started, in progress, completed).
- Cancel rides with notification to all passengers.
- Passengers can cancel participation.

8. Admin Functions

- Monitor system usage.
- Handle dispute resolution.
- Generate reports on carpool statistics.
- Moderate user content and behavior.

QA-Centric User Stories (For Automation)

These user stories are designed to be testable using code automation tools.

User Story 1: Creating a Carpool Ride

As an Employee with a vehicle

I want to create a carpool ride from my home to the office or from office to home

So that I can share my commute with colleagues.

Acceptance Criteria:

- Given I am logged in with a verified account
- And I have registered my vehicle details
- When I create a ride from “123 Home St” to “Office HQ” at 8:00 AM with 3 available seats (or from “Office HQ” to “123 Home St” at 5:00 PM)
- Then the ride should be created successfully
- And it should appear in the available rides list
- And I should receive a confirmation notification

User Story 2: Joining an Existing Ride

As an Employee without a vehicle

I want to find and request to join a carpool ride

So that I can commute to work efficiently.

Acceptance Criteria:

- Given I am logged in with a verified account
- When I search for rides near my location “456 Home Ave” going to “Office HQ” around 8:00 AM (or from “Office HQ” to near “456 Home Ave” around 5:00 PM)
- And I select an available ride and request to join with my pickup location
- Then the ride owner should receive my join request

- And my request status should show as “Pending”

User Story 3: Approving Ride Requests

As a Ride Owner

I want to review and approve join requests

So that I can control who joins my carpool.

Acceptance Criteria:

- Given I have created a carpool ride with 3 available seats
- When I receive a join request from employee “Jane Doe”
- And I approve the request
- Then Jane should receive an approval notification
- And the available seats should decrease to 2
- And Jane should appear in my passenger list

User Story 4: Seat Availability Restriction

As the System

I want to prevent requests when no seats are available

So that rides don’t get overbooked.

Acceptance Criteria:

- Given a carpool ride with 0 available seats
- When an employee tries to request to join this ride
- Then they should see a message “No seats available”
- And they should not be able to submit a join request

User Story 5: Route Visualization

As an Employee

I want to see the route of a potential carpool on a map

So that I can determine if it works for my commute.

Acceptance Criteria:

- Given I am viewing details of a carpool ride
- When I click on “View Route”
- Then a map should display showing the route from start to destination
- And my location should be shown in relation to the route
- And estimated pickup time should be calculated based on my location

User Story 6: Ride Reminders

As a Carpool Participant (driver or passenger)

I want to receive timely reminders about upcoming rides

So that I don’t forget my carpooling commitments.

Acceptance Criteria:

- Given I am part of a carpool ride scheduled for tomorrow
- When the system time is 24 hours before departure
- Then I should receive a reminder notification
- And another reminder should be sent 1 hour before departure